SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

BUS105

Timed Online Assignment - July Semester 2022

Statistics

Thursday, 17 November 2022 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **4** questions and comprises **7** pages (including cover page).
2. You must answer **4** questions.
3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.
4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**
5. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. **Images of handwritten answers will not be marked**.
6. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.
7. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

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(Full marks: 100)

You must answer ALL the questions. **(Total 100 marks)**

Question 1

Recent news showed that couples get married late in Singapore, and the median ages of first-time grooms and brides were 30.4 and 28.8 in 2019, older than 29.8 and 27.3 in 2008. Jerry, a student in Singapore University of Social Sciences, wants to examine whether the marriage trend is seen among his friends. So he asked around and collected some data on the age of first marriage of his generation and his parents' generation.

Table 1-1 shows descriptive statistics for the two groups of data. ***Figure 1-1*** and ***Figure 1-2*** present histograms.

	Age at First Marriage	
	Young generation	*Old generation*
Mean	27.941	25.345
Standard Error	0.539	0.388
Median	28	25
Mode	30	26
Standard Deviation	2.221	2.092
Sample Variance	4.934	4.377
Kurtosis	-0.784	0.946
Skewness	-0.149	0.009
Range	8	9
Minimum	24	21
Maximum	32	30
Sum	475	735
Count	17	29

Table 1-1

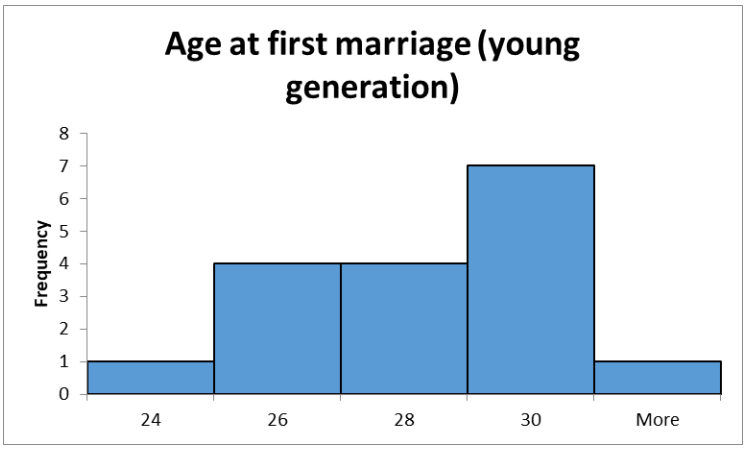

Figure 1-1

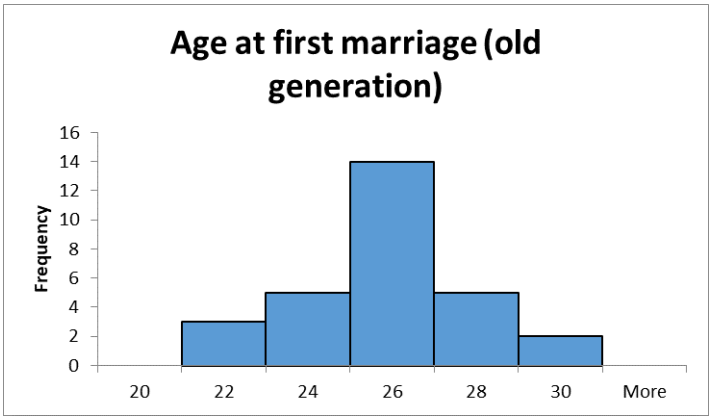

Figure 1-2

(a) Describe ***three (3)*** measures of location and ***three (3)*** measures of dispersion. State the values of these measures for both young generation and old generation.

(9 marks)

(b) Interpret the results presented and report them as an executive summary. You should include comments on the central tendency, dispersion and histograms, and examine whether the trend of Singapore couples getting married late is observed in the data.

(10 marks)

(c) Explain ***two (2)*** concerns in his statistical analysis for the marriage trend.

(6 marks)

(25 marks)

Question 2

(a) 120 students were surveyed on their preference on the course delivery mode. The results are summarized below.

	Preference on the course delivery mode		
	Face-to-face	Online	No preference
Male	21	27	6
Female	28	25	13

Table 2-1

Using the 120 students as the sample, a student is randomly selected.

(i) Identify the probability of selecting a student whose preferred course delivery mode is either face-to-face or online.

(ii) Identify the probability of selecting a female who prefers online class.

(iii) If a student selected has no preference on the course delivery mode, what is the likelihood that the student is a male?

(iv) If the student selected is a female, what is the probability that the student prefers online class?

(10 marks)

(b) According to a government study, the average monthly household income in Singapore is $12,235 in 2020. Assume that the distribution of the monthly household income follows the normal distribution with a standard deviation of $5,600.

(i) What percent of households have a monthly income less than $14,000 in 2020?

(ii) What percent of households have a monthly income between $5,000 and $10,000 in 2020?

(iii) 13% of households have a monthly income less than V dollars in 2020, report the value of V?

(iv) If 10 households were randomly selected in Singapore in 2020, what is the likelihood that the average monthly income of the 10 selected households is more than $10,000?

(15 marks)

(25 marks)

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Question 3

When there are increasing cases of locally transmitted COVID-19 infection, dine-in ban is used as a measure to control the spread. To study whether the number of times a household buys take-out meals in the month of dine-in ban in the eastern and western regions is similar, a research team randomly selected 32 households in the east and 40 households in the west.

The summary statistics for the two sets of data are given in ***Table 3-1***, and the test results are given in ***Table 3-2***.

Take-out Meals in the Month

	East	*West*
Mean	29.344	30.550
Standard Error	3.119	2.632
Median	25	24.5
Mode	20	13
Standard Deviation	17.646	16.647
Sample Variance	311.394	277.126
Kurtosis	-1.240	-1.316
Skewness	0.305	0.341
Range	56	52
Minimum	4	8
Maximum	60	60
Sum	939	1222
Count	32	40
Confidence Level(99.0%)	8.560	7.128

Table 3-1

	East	*West*
Mean	29.344	30.550
Variance	311.394	277.126
Observations	32	40
Pooled Variance	292.302	
Hypothesized Mean Difference	0	
df	70	
t Stat	-0.2975	
P(T<=t) one-tail	0.3835	
t Critical one-tail	2.3808	
P(T<=t) two-tail	0.7670	
t Critical two-tail	2.6479	

Table 3-2

(a) Calculate the 99% confidence intervals for the mean number of times a household buys take-out meals in the dine-in ban month in the eastern and western regions, respectively. Interpret the confidence intervals and discuss your findings.

(8 marks)

(b) Apply an appropriate hypothesis test at the 1% significant level to determine whether the number of times a household buys take-out meals in the eastern and western regions is similar. You are required to write down the ***five steps*** carefully.

(10 marks)

(c) Compare the results from ***part (a)*** and ***part (b)***. Should the conclusion be different or similar? Please provide your justification.

(7 marks)

(25 marks)

Question 4

Mr. Tan is a market researcher who is currently studying factors affecting the number of subscriptions for online services. He found the following information on 50 online subscription services:

SUB: Number of subscriptions (thousands)
HIT: Average monthly web page hits (thousands)
ADV: Advertising budget of the online service ($hundreds)
PRICE: Monthly subscription price ($)
Multiple linear regression is used for the analysis, and the regression output is shown in ***Table 4-1***.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.543
R Square	0.294
Adjusted R Square	0.248
Standard Error	0.806
Observations	50

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	12.467	4.156	6.398	0.001
Residual	46	29.879	0.650		
Total	49	42.347			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	29.157	2.669	10.923	0.000	23.783	34.530
HIT	0.005	0.003	1.593	0.118	-0.001	0.011
ADV	0.058	0.024	2.472	0.017	0.011	0.106
PRICE	-0.145	0.046	-3.128	0.003	-0.239	-0.052

Table 4-1

(a) Describe the relationship between the number of subscriptions and the ***three (3)*** independent variables by writing down the linear equation. Interpret the coefficients to show how each of the independent variables impacts the number of subscriptions.

(6 marks)

(b) Mr. Tan wants to estimate the number of subscriptions for a particular online service which has 720,000 web page hits per month on average. This online service has a budget of $2,500 for advertising and intends to set the subscription price at $30 per month. What is your estimate of the number of subscriptions?

(5 marks)

(c) State and interpret the adjusted coefficient of multiple determination. Explain its implication in this case.

(5 marks)

(d) Execute appropriate hypothesis tests to advise Mr Tan which variable is significant at the 5% significance level. You are required to write down your steps carefully.

(9 marks)

(25 marks)

----- END OF PAPER -----

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